ActInf Livestream #027.2 ~ "Active Inference: Applicability to Different Typ

Livestream | 1:52:04

hello everyone and welcome to the active
inference lab this is active lab live
stream number 27.2 on august 31st 2021
we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us
at the links here
this is a recorded and an archived live
stream so please provide us with
feedback so that we can improve our work
all backgrounds and perspectives are
welcome here and we'll follow video
etiquette for live streams and see who
joins us
the short link is the calendar of live
streams
of the past and future and we did choose
28
paper
but didn't update this slide so check
the link for next week's paper
today in 27.2
welcome to
stephen greetings in 27.2 we're
joined by the two stevens as it were
and um we'll
have a few questions that we can
have slides for the figures to go
through the paper itself anything else
that anyone wants to add and then of
course
any questions that
people type into live chat as well if
they're watching live so
um we can just

say
hello and then we can
[Music]
leave it to the author last just to
maybe
open it up or just bring what they
were thinking about
our second discussion here today so i'm
daniel and i'm a postdoc in california
i'll pass to stephen celet
oh hello there i'm stephen i'm based in
toronto and uh
i'm working on social topographies and
other spatial meaning making and hoping
to find ways for active influence to
shed light and all of that
and i'll pass it shall pass it to the
other stephen is that okay
yeah good day to you i'm uh i'm in
finland southern
finland i'm based at the technical
research center of finland and i'm
ready to answer questions
cool so again
uh anyone who is watching can post a
question but then there's a few that we
have lined up a little bit
just one
theme that
stood out
as far as selecting this paper was that
it was applied it was applied to making
the kinds of objects and
supporting
[Music]
the ability to have the internet and
computers as well as just objects like
pens and things like that so

how
does this idea of helping us design
physical systems which ends up being our
niche modification
uh
activity as an economy
how do we help design maybe if we're not
an industrial engineer
but potentially people working on
different kinds of projects that end up
modifying the environment
maybe even influencing the same systems
that industrial engineers use
or um using some of the same techniques
but people who are just thinking along
these lines and want to take what you
translated one step towards application
in your paper and then think about how
it applies in their own
work
so cyber physical system so the latest
um
trend in this
is um
for example the in
in the usa i think it's called smart
manufacturing
and in europe industry 4.0
and a big thing in this is the idea of
having digital twins
or digital siblings
or even
avatars um
and the idea of this is is that
there's a
the physical
machine
the physical process has got um

sensors on it and actuators on it
and the sensors
bring live
uh
data
to the digital equivalent
of this physical process
and through
so uh
and then it seeks to optimize by seeing
what's going on
so
and a basic
um
basic level is although difficult to
achieve is predictive maintenance for
example so if nothing else at least to
stop the physical process from
malfunctioning or even worse stopping
and there's no um
[Music]
there's no
[Music]
theoretical
um
foundation for
for these industrial
innovations for things like digital
twins so of course there's a theoretical
and background of cyber physical systems
but there's nothing to
anchor it with in my opinion
that's very
straightforward
so if we look at
life sciences generally
we know that
evolution has had

millennia of
and millions of years to iterate
to
develop elegant solutions
so it seems sensible at least to me to
to look to the life sciences
to see how nature's evolved towards
least action
and
of course uh
free energy principle and active
inference
framework or theory
uh
[Music]
from the life sciences
and
are
in their essence i believe very
straightforward
as one would expect guiding principles
in nature to be
and applicable to
human and artificial intelligence to
artificial agents
so
i believe it's useful to start from that
basis
and with
the free energy principle
this very straightforward idea of that
of trying to minimize
the information gap
between
the generative model
and the world itself
so that's a very straight forward idea
and i know there's

um you know the
the mass behind it is evolving all the
time and there's some debate and
uh
courteous dispute about that but the
basic ideas seem straightforward so
i believe it's therefore a good place to
start
and
because this is such a basic idea isn't
it
minimizing the information gap between
the the generative model and the world
itself
is this what i've said helpful in any
way
thanks for this i wrote a lot of it down
stephen's let
yeah some really good points there
one thing i'd be interested in to to
know what your thoughts are is
you know we can end up kind of
adapting our niche through action as a
kind of consequence
of
like following a path and these types of
things
and then we've got this kind of more
constructed um
sort of uh
deliberative um
organizational work
um in which you know the in many ways
the point of that work can be to
construct the environment um
in which tools will operate you know so
in some ways we we use the tools of our
legs or the tools of being in

in the world and it can
sculpt that niche and also we can be
very much sculpting a
bounded niche
so i'm wondering my question is
thinking about
whether you see those that kind of
distinction as being significant and
how you see um
the
idea of like the free energy or the
implications of the free energy sort of
rolling out in the niche versus
the generative model for that niche
being embodied in some way so you know
so anyway so how much is there something
when it's embodied
for instance when you're talking about
these um
total quality management environments
where
someone is um
trying to maintain
i don't know iso standards of
manufacturing or whatever so
i wonder what your thoughts are about
that whether there's distinctions there
in terms of active influence being
particularly involved or
you know the free energy principle more
more generally
so um
so i believe that um
that that the core of quality management
is
just a
an instantiation of
free uh of

realizing the free energy principle
through active inference
and that's what i've you know put in the
paper
um but there's a bigger issue about um
[Music]
that in in nature things are evolving
and uh
as i put in another one of my papers in
entropy that uh
that
so
there are no
things are evolving to some degree
naturally and
so if you say predator and prey
they they just it's evolved and if
you're a beaver trying to get back to
the
lodge that you've built your niche
construction as a beaver
and there's a links in if you're if you
see in between
where you are and you'll be the lodge
and there's a links
a hungry link standing in your way
then it's clear that okay and this is a
predator and i should
this is not my as a as a beaver that's
not my preferred observation
whereas my uh beaver lodge is a
preferred observation
but what goes on in uh
human niche construction and uh because
it's the main area environmental
engineering
is not so straightforward so
um for example

highly refined food
it seems to be a
naturally to us like a preferred
observation because it tastes good and
so it makes life less strenuous and more
stimulating
so if they're quite if the if the if the
company
making
uh
highly processed food
ready meals
and so on
it's it's acting within
quality
assurance so it's doing in my opinion
active inference
to
realize the free energy principle
but
what its output
is
it subverts the free energy principle
the externality it subverts the free
energy principle because
what it's offering to us as consumers
it seems to be
consistent with
what revolved to do so it makes life
less strenuous
and more more stimulating
but actually
it's not good for our health in the long
term of course
now and then no problem
but if that's all you if you live in a
food desert
if you live in a food desert we know

there are lots of studies that show that
unfortunately
this isn't good for health
so if you take
one
um
entity at a time
it can be
conforming with the three energy
principle um through active inference
but
its output
subverts
the free energy principle because it
makes
our profit our
our preferred observations
instead of being good for us
are
not good for us but we only discover
that
20 30 years later
so that's quite a lot to i have a
diagram in um
uh
there's a simple diagram of this in
[Music]
get paper
so the paper's called
yeah it's called multi-scale free energy
and analysis of human ecosystem
engineering
okay
and there's uh
one diagram in it figure one
so it's an entropy as well yeah
shall i share my screen
sure

now remind me there it is isn't it
okay
all right
are you saying
multiscoping free energy analysis if you
make your system engineering yep
all right so
here's this diagram then um
there it is
so what
what the so what active inference is
leading us to is that
from nature
is that our preferred observation
supports survival
an unwanted surprise so
an unexpected difference
between
uh
a prediction
of what we're going to experience based
on our internal generative model so an
unwanted surprise so if the beaver's
going back to its lodge it's thinking
it's going to have an unhindered path
back to its lodge and there's the
unwanted surprise of the hungry lynx
standing in the way and it threatens
survival
so that's natural free energy principle
but
through um
human
environmental engineering
niche construction within that
preferred observations
can threaten survival
so if my preferred observation is

a comfy sofa
a large
tv screen
um
and a
device to change
channels and do uh gaming
and then i can just order some food in
it comes
uh very tasty but
perhaps not good for me i could be
sitting down for
16 hours and then go
go lie down for the remaining eight
hours
maybe all my preferred observations but
they're not good for me
whereas the unwanted surprise that
supports survival could be
messages that okay we should not sit
down for more than 20 minutes we should
take
lots of physical activity we should eat
um
make our own food
and so on
so
what's going on in human endeavors
which
may
individually
conform to realization the free energy
principle through
active inference can end up
subverting
the free energy principle
outside
of that productive act of that

production activity
and here there's
some numbers
okay so
so beaver could develop a new genitive
all right let's see here
all right so so in nature
for a beaver the
the um preference distribution for
sensory inputs is for example plus 10
for the protective pond the lodge
zero for woodland and minus 10 for a
hungry predator
but then if there's a bit of pollution
you change like okay so now it is
getting more complicated so there's
plus 10 productive bonding clear water
plus two predicted upon polluted water
woodland zero hungry predators still
minus 10
but then um
uh
or
the example in this paper is in um
some uh island nations very small island
nations where
there's um
a lot of important cheap calorie rich
nutrient poor food
and then the preference distribution
could be plus 10 imported food plus two
local food minus 10 no food
even though there's clear evidence that
the imported food is
not good
okay
so there it is
[Music]

okay so did that um
did that
lit up the hands
i i wrote down some stuff as well but
blue welcome um
go for it and then
afterwards we'll be stephen
thanks so i i was really curious about
that subverted um
model and like where
you're maybe not optimizing like where
where what your preference is is not
optimizing for survival
and i think that um you know you used it
in the human context but i can think of
many contexts where animals do a similar
thing like a moth will fly into a flame
and like a cat like i have these two
kittens now in my house and they're like
such trouble right like their curiosity
like had my cat trapped in a box earlier
today like i mean trapped like under
furniture in the box which was like so
bizarre so i mean i can think of of
models where it's always not optimal and
so is this something that you think
excuse me what was the last thing you
said
it just something that you frame as
uniquely human because i can think of
many instances of like non-human
uh creatures that also have less than
optimal preferences
um no i'm not framing is only human but
what i'm
pointing out is is that
so for example the moth flies towards
the flame

but where did the flame come from
and the cat's curiosity
luckily did not kill your cats that's
good
um
but they were going into boxes
but where did the boxes come from
so
are these things
um
of course there are misjudgments
so all living things can
they could be uh
and of course with predators and prey
that for sure the predators are trying
to get the prey aren't they
and people can have um and
living things can misjudge the how far
away the prey is and so on and signaling
systems can
can get messed up and things but
with
human beings the scale
and the determination
of
um
activities
is is much
larger so
there's a
there's an american author who's his
latest book i mean he's called hooked
um
but uh he's studied the the for example
the food industry for many years and um
he
his
view is that

there's nothing at all malicious going
on but it's just if you're
if you're in the in the uh
a certain
business
you want
you're trying to get as many customers
as possible
and you're trying to retain them
and
um
with no no bad intentions at all it's
all good intentions
but
then the effects
are
we discover it can be decades later that
okay
um
this is
what we thought was
good
is not so good
whereas in nature if the
the wherever the flame
suppose it was a flame from somewhere in
nature it was a natural fire
well the
the natural fire maybe came from bolt of
lightning or something
it
wasn't a determined effort over many
decades
of ever increasing refinement
of a market offering
but uh so
there's more
so what i'm showing in the paper that

you read
that diagram
about how the quality management system
is like a general it's like an active
inference system
to reduce
free free energy
so
it's optimum for the organization
but it can be sub-optimal
outside the organization so i think a
general word to them for this is
externalities
just
uh two quick things one was trying to
explore that distinction between what is
human other than the fact that we care
about us and each other
what is human and i was just thinking of
the ants foraging out there in the
desert
and they don't do margin trading
or um
staking of their seeds to try to get
fractional seeds and lending of seeds
it's like layer one seed gathering
versus
the higher order
symbolic risks that the humans engage in
so that and the niche modification and
the scale among other factors and then
also what you said about misjudgment and
the ability to misjudge because
a lot of people think oh well you know
if um let's just say
traditional understandings of evolution
are about fitness
then

why why do organisms fail
why do all organisms not just maximize
their fitness or if active inference is
about reduction of uncertainty or
using action and inference to reduce
uncertainty then why do organisms fail
and that's where the engineering
perspective comes into play like
things just do
fall apart and just because there is an
imperative whether to maximize fitness
or to minimize uncertainty the point is
of
the ensemble there's going to be a
distribution of performance like 100
engines they're going to have some range
of failure times the ld50 for some
pharmacological experiment so just
because the framework has sort of a um
imperative weather fitness or
uncertainty reduction
doesn't mean that it's utopian or like
pangolossian it means that every single
thing always works perfectly
so just
cool points that you raised
but it but also blue um my my uh
i have imperfect knowledge that's for
sure
and i wouldn't make um
i i i i don't know enough
and maybe nobody does um like
what
was just said about ants of course
those words were coming for an expert
and i don't i know next to nothing about
that and
many of the behavior of many of or

animals i don't know so for sure it's
not
it's not clear-cut it's not you know one
that humans doing this and all other
animals aren't doing it
thanks steven celet and then of course
anyone who's watching live
i think this raises some useful
areas where we have spaces where it's
very free energy minimizing as the main
kind of
way of getting a handle on things and
the broader active inference so for
instance
when we're dealing with equilibrium
chemistry we're dealing with
things which are
often boiled or heated or
made to be
effectively dead
which is what happens in most
engineering processes in most food
processes i mean the main thing that
coca-cola
does by putting sugar in is it makes a
preservative
someone's saying to me oh why can't they
just use water it's an awful lot easier
to distribute coca-cola than to
distribute bottled water because
coca-cola doesn't go off because it's
got so much acid and sugar in it it
won't go off right
so that what you've got is you've got
these processes like you're talking
about with tqm
and i worked in the pigment
manufacturing industry so we they were

bringing in the iso standards there
around
pigment manufacturing because they were
inheriting a lot of these formulas from
like 100 years previous
and it was all about
minimizing and to make things
reproducible and within tolerance
but you're always dealing with
you never want things to
be organic you don't want things to sort
of
to grow
in any place in the in the process you
want it to all be
i suppose you could say dead right but
then what's really interesting as you
point out is
okay that might be what happens within
the containment of the factory or the
industrial process but it will go back
into the world just like it will when
it's out now as a product
it will be used in
an ecological niche and it will not be a
dead niche you know that food that maybe
isn't even biologically feasible or some
of those chemicals are not there's no
biological route to create them there's
no way to create fluoride
compounds biologically it just can't be
done it has to only be done
synthetically
so what's your thoughts about that
sticking with minimizing free energy
more when you've only got dead
equilibrium systems and then when it and
what and how much active inference is

involved in in that i suppose a first
order is only a second or third order
and then
what how things pop out into the niche
in the community or whatever
um i think this is um
open question i mean at one level maybe
it's um in world systems theory this uh
the the framing is that there are so now
it's in about whole countries rather
than just um a company making something
in one factory that
there are inevitably
core i think the terminology is core
countries that
so it was the netherlands then britain
then the usa now going towards china
and then there are
peripheral
countries
two levels of those and um
the argument is that the the development
of the core countries is
dependent upon taking from the
the peripheral countries and
um
is this
entropy being shipped out of the uh
so if if you think of like a a factory
as a heat pump
and it's uh
you're getting all the entropy out of
the factory operations
but then it's
where's it going
is it being shipped out of the factory
and then
it is entry but you've been shipped out

of the
pumped out of the core
countries into the rest of the world
where things are confusing so
i don't know i mean that's a bit of a a
broad sweeping statement isn't it so but
maybe
maybe on one level that's what's going
on if there's
if you
if you have supply chains
value chains
and
you push all the physical disorder
out of those by eliminating all the
information
uncertainty then
does that mean that the physical
disorder has inevitably got to go
somewhere else
i don't i don't know
so uh but
maybe and then
but uh
with
but for sure there are externalities and
the interaction between levels is uh
because in inhuman endeavors it's uh
it's a shifting matrix it's not just a
simple hierarchy
can i just ask one thing just bouncing
back off that is i think it's an
interesting so in some ways
by
by things sort of being taken out of the
ecological niche effectively by putting
in a process a processing plant where
it's effectively dead you know

you you heat up your sugar your sugar
cane raw material in a vat with sulfuric
acid and it
you know it's all it's all dead it can't
it's basically a chemical process that
gets put into
whatever foods that someone's eating
the consequence of having that
dead
equilibrium based process rather
non-equilibrium steady state process is
that you're saying it puts a burden back
on
the ecosystem to try and reintegrate it
so in some ways
is it pushing entropy out or is it just
placing a burden on
system to try and find a way to cope
with something that it can't
naturally incorporate easily into its
niche
what do you think blue
yeah i'm not sure really i'm not um
i'm not uh entirely certain
on this
on this um
transfer of disorder
it's definitely very important like for
example when people say well look at the
miles per gallon of this car and then
okay but where did the metal come from
okay but you know everything is tied to
everything and how we evaluate
policies ultimately and choices are
critical and we do have to understand
the relationships of um
the world's supply chains but then i
thought well it's not like there's a

like we're in a bomb calorimeter or a
heat bath where there's a constant
um number or proportion of some kind of
uh negative activity like there could be
an ant colony with
of in a niche with very low
nest mate aggression or bodies like
there's organisms with low rates of
cancer or in certain niches you know
it's not like a statement of them across
all
patterns and times
but
um but then again it was like so things
do fail
and there's no perfect
state
but then at the same
time
it's not zero sum in the informational
game it's not like we're trying to
because we're getting energy coming in
so can we kind of transmute
physical
order with information disorder
do we have to like offload something to
another place physically
does it bump up against any of the
information
limits or calculations or density
of processing or storage
will we not be able to remember enough
and then there's a catastrophic failure
i mean there's so many
components to your questions so i hope
that we can all keep thinking about that
because it's important at the level of
like

human
activity with the supply chain and
that's also an important long range
question
and if we don't have a long uh range
preference
then we're not going to get there
can i have one more bit in there without
yeah
um
i think what's interesting as well is
there's this question of
because i don't really believe there is
strictly speaking neg entropy as such so
in terms of
um it's always kind of a relative thing
but it's a useful it can be useful to
think about it that way sometimes but
ultimately it's always a change so i
suppose the question
comes in in terms of
how much is it around the ability to
infer
using entropy
as much as it is about
how you know how much is it to be able
to use change in entropy or dynamics of
entropy which can be a complex jungle
niche you know it could be a lot of
entropy change and a lot of stuff going
on but it can be very healthy
how much is that ability
to
dynamically get at the variational free
energy really important for life
um whereas for us it's all about you
know at times in an engineering context
it's just reducing free energy per se

as a kind of an overall entropy term
which could be actually not so much
about variational change it's just an
overall general
reduction in entropy i wonder what
what your thoughts are on that
one thought on that on the biological
side is
what's an industrial engineering
relationship ecosystem engineering
relationship that has persisted
for billions of years or millions of
years or just long time however you want
to see it
the nitrogen cycle and some of the redox
cycles in the bacterial
uh
relationships biofilms the mitochondria
and the eukaryotes
now that's a shifting interface
but on the other hand eukaryotes can
totally count on that interface
the powerhouse of the cell as they say
to persist
but then there's this other sense
in which there is still a game theory
and a failure mode and uh probably an
interface of maximal uncertainty
and the requirement for active sensing
and trade-off costs
so it's again it's not like things i
think
work 99 of time or 100 or 20 percent of
the time
that is kind of how the chess board is
set up for the
for the
um

for it to even be a winning board at all
there's just niches where it can't work
like there's temperatures that just the
population will just die at period
so is our engineering condition such
that there is a path
that's going to lead to
what standard of life
with what precision and
uncertainty
for what distribution of people
like did we way way overshoot
and now it's going to change back down
to a different way or
is there some critical path
to
changing
the distribution of industry
because it's not going to be
any way it was from
zero to
2021 it's not going to be that way next
year
um
are you
do you
are you still seeing my screen oh
i'm not could you reshare it
okay all right this is a
can i reach
you do that again
this is a are you seeing now
figure one better ecosystems in the
anthropocene
now i see it yep
okay yes this is a paper i'll go to the
title um
so

nearly awesome or anything but
this is um
uh not by me by others nothing to do
with me i'm just the reader of it um
i know one
thing that they're saying that
you can just
so here this
this is the same
these are showing the same things but
a is where human impacts on
migration-based matter ecosystem
it's
it's just sort of implicit
and on this framing of it they say that
okay so we explicitly deal with
the uh socio-ecological
framework about okay so what
are human beings
what mental models of human beings got
what regulations arise from that and
then
that inputs into it
so let me go to the top so you see the
title
yeah embedding method
on embedding meta ecosystems into a
socio-ecological framework
and this is a response this is just a
short paper it's a response to somebody
who
replied to the no it isn't actually
nope
i'm mixing up with another one there it
is that's the title anyway
so i'm
wondering if um
that may be that

each each way of um
uh
making things
has got its own
entropy dynamics
and the in this entropy and dynamics
that there's
physical disorder is reduced
information uncertainty is reduced
but maybe that then pushes
the entropy
out
elsewhere
and then just doing more
of the same thing it's just not possible
there's a limit to it
and with world um systems theory
uh
this is the breaking gist of it that
there are these core countries and there
are peripheral countries that serve them
and
maybe the if there's one
way of doing things
you can't address
the entropy that's caused by that
by trying to do more of the same
or trying to tweak
that way of doing things
and maybe something
quite different has to be done
and maybe there's something about
something to be learned from
landscape restoration ecology
in this about that
about um habitat fragmentation
that that
okay so here are these habitats and

they've grown and
they're thriving but these ones over
here they fragmented they split and this
there's nothing from them but dispersal
so as people are dispersing from
um
parts of the world that are
suffering from premature
deindustrialization
that
maybe there's something in
entry by dynamics that
could be that well
that this
doing this type of industrial production
this it drives out the entropy but it's
shipping it out it's going
out elsewhere
and the entropy that is then um
in other places it can't be dealt with
other than by some in
another way
of existing
does that sound in any way sensible
a very
interesting area
multiple
levels of nested modeling and
uncertainty and how informational and
thermodynamic
uh
aspects
transmute into one another and arbitrage
with one another or
relate to each other so it's really
interesting area
and
uh

one kind of thought on that and again to
the fast food and things that might be
unhealthy for us or policies that might
be unhealthy in the long term for an
organization
it it made me think about
being on a road trip
and knowing that there was going to be
not just the hyper stimulating
experience
which has been argued in many other
frameworks like in reward based
frameworks which is what you know
dopamine the reward molecule
reinforcement learning reward learning
they'll
already have a framing for how those
products hyper stimulating lead to
failure modes of complex societies via
hyperstimulation okay so that is sort of
a
common stance
with active inference we can also kind
of highlight the fact that those
products have lower variability
and that there's actually very bland
foods
that for reason of blandness that people
or like a tv show or something that for
reasons of narrative or
aesthetic
just um similarity people will re-watch
i mean even to videos that of types of
people didn't expect
why do people watch the log burning or
the goop playing
because they're sort of like the um
frontier of optimal foraging is about

like how it's going to sound which is a
very small thing to be resolving
relative to the more um
deeper unsettling levels of information
that could be revealed like someone
else's
perspective of you or something like
that so it's just interesting how we can
actually
as you point out like the supply chains
are modeled on one hand
by these cyber physical
digital twin
um
frameworks that
need a unifying
approach and so on one hand active
inference does provide that but then
looking at the other side of the
equation so what you didn't discuss in
our paper for these weeks but where you
left off when you start introducing the
biological
and the survival imperatives
then we can also have some new ways of
thinking about how those industrial
products
through precision hacking rather than
reward hacking end up leading to
maladaptive behavior with or without
reward incentivization
are you um what are you seeing on are
you seeing my screen now a different yep
very different diagrams
you see birds yeah generalist bird gap
specialist butterfly
interior habitat but especially
speedless so this is another paper

um i'll again i'll scroll up to the top
you see this tile
um and it's making this point that uh
landscape composition and configuration
go to different species perceptions
and so the first one's a human and then
it's
different species
and as it's neatly illustrated there
that the same the same place looks
different
to different species and
so what i'm
and
the the kind of
the terminology here is that there are
patches
habitat patches and they can be
corridors between them
or stepping stones between them
and the stuff the other stuff
is called matrix
and i personally can't
ever see that word or say it without
thinking about the movie so
that is a bit of a problem but uh but in
the landscape ecology is called the
matrix so
this i think this could be useful
of illustrating that well
okay that so
here's the road to the mine these uh
front country come
companies built the road to the mine
they built the mine and they're taking
the
stuff out and they're taking it back to
their country to

[Music]
do highly efficient manufacturing with
it and then they send their completed
goods back and
but then what's left
um
so if you just measure that in terms of
gdp growth then the gdp goes up divided
by the number of people and it appears
that prosperity is
rising but
if
if you look at it in this kind of view
it shows that well there's a lot of
blank spaces left to be left here
and uh
if we think about it in terms of this uh
restoration or regeneration
it
it
it's looking at it all of these because
of course the the perspective of the
human the generalist bird the gap
specialist butterfly and
the beetle they're all equally valid
and um
so
you know what this paper's called
landscape ecology and restoration
processes
and even within the same physical
overlap
one bug might be underground or in the
roots or on the ground in the canopy so
even the physical micro niche
that different strategies or individuals
experience in the same
spot in the matrix can be very different

so there's just such
richness to this ecological thinking
and
it's just
challenging to bring it into play in
some of our cognitive endeavors but i
think that is also changing rapidly
so i've taken um
[Music]
an ecological perspective
in
we're doing one paper on meme ecosystems
and rhetoric ecosystems
by the end of this year
but it's not active
direct
but
maybe it'll have some
relationships it could well do um
are you now seeing a table to number one
all right so what i've done here is i've
done this from a uh ecological
perspective and
well you can see these terms but one
thing in uh
a
restoration ecosystem restoration is
mobile linking organisms
but
that like birds and animals that can
travel quite a long way they can go so
after this is a forest fire
then
animals that can travel quite a long way
particularly birds they can come they
can fly and drop seeds and and things
like that so this is an established
concept mobile linking organisms and i'm

pointing out that this is something that
we've done in mobile factories
and then energetics another key issue is
that if there's no
if all the energy if all the matter and
energy are flowing
out
then life
isn't sustainable
and uh of course there we are so i've
i've done that
relating to my uh
favorite subject movable production
technologies
okay
[Music]
so
um
so that takes i think this is
decentralized systems isn't it that
uh
the second point
that was it that was the lead in the
segway okay nice
how about oh go for it um stephen
yeah so i think this is the
if everything's so at the moment there's
this uh
centralization oh let's have a look at
this i've got um
i go again so
let's have a look here
are you
what do you see now do you see a
taxonomy
uh taxonomy and manufacturing
distributions and their comparative
relations to sustainability

all right so
here they are
so
there are distributed distributions
and centralized
um
and as we see there's plenty of
distributed options
and uh then i compare them in
terms of economic ecological social and
institutional sustainability
and where i think
active inference i've just
started to
look into this
that this agent environment systems
and uh
at framing is
the the agent has a generative model
but the world is a generative process
the sensory inputs are coming from the
generative process to the world
and um
i
think that this enables um
active inference to be
linked to
uh
ecosystem dynamics
and
[Music]
that
that what we've been well what i've been
saying before about this centralized
distribution it um
it's super
efficient
and it's getting uh engineering out

physical disorder and information
uncertainty but
then
maybe that entropy is going somewhere
else and uh
maybe
new information is needed to activate
the more distributed
options
so do you think do you think that um
that
if
if the only if only a certain um
so if
if there's a predominant paradigm which
in this case there is industrial
manufacturing
based on the 300 years old paradigm
um
that
the information about how to make things
is all about that and how to improve
that
and
then
then does it become a zero-sum game
that
if if
so if it is shipping out um
entropy like a heat pump
out to the peripheral nations do you
think it's a zero-sum game if there is
an alternative information
so if nobody knows
or there hasn't yet been
[Music]
uh conceived
an alternative

then then is it like a closed system
so in a closed system
the entropy
has
it's
if it's pushed out of one place
it goes to the other places
so is it is a cl is it a closed system
is it a closed system
if all the information
is about one
paradigm so is this like uh for example
does am i in any way coherent here
it's an extremely important
question
and um you know
socio-technical systems political
systems tokenomics that last forever
are these the the perpetual motion
machines
of the 2020s
and just a new
level
of uh computational
uh detail and flourish
but the same kinds of
fallacies and then
whatever the thermodynamics of the earth
are
our niche is finite and we've now made
some potentially
um
[Music]
whether or not the sum is zero or even
negative
or could be positive but is negative
there may be
informational and strategic

games that um
don't resolve like heat baths
so
they can't be understood
with some of the
at least most
tangible
physical
metaphors that thankfully guide
the evolution of
theories that our first principle like
fep like carl
fristen with the wood lice
seeing the trajectory of a real thing
and that's based upon health and um
longevity
blue
yeah i had a question about the table
that you pulled up earlier um
specifically about ecosystem engineering
and so i'm not sure if it's just like a
term that i'm not familiar with or like
in in terms of engineering i'm kind of
naive
so
when you're thinking about ecosystem
engineering um i i i equate that to
niche modification or like designing a
system like even you know you could
design like a world system
so
here it says it does not require
overriding the natural evolutionary
balance of equal equal ecological
fitness but like i think about like the
human paradigm blue could i just could
it just
happen

but that's movable production
technologies
so what i'm
saying there is that
if you've got
a movable factory so a truck if you've
got a secondhand truck
with a secondhand container on the back
of it
it does not require overriding the
natural evolutionary balance
so
it's it's specifically in relation to if
you look at the top it says movable
production technologies
so so how is that ecosystem engineering
though like i mean no no it isn't yet
but the point uh obviously this table is
not clear
and it is my responsibility it's not
clear
i am responsible for this but
so the the
what this meant by this is because
further up in the paper
it's pointing out that if you have fixed
production technologies
so if you've got a big fixed factory
you build a load of roads huge roads
going to it and so on that involves a
great deal of ecosystem engineering
but
a mob a mobile factory it can just go
over rough terrain
also to uh play devil's advocate against
the author
the enabling function
of

a uh realization of something something
physical that can be an ecosystem
engineering agent ak niche modifying
agent um that it's gonna have to be a
niche modifying agent so this is just
sort of
then connecting that to mobile organisms
so you have quote plants
aka factories that are planted
and then those can be like
they're not like trees simply but
they're planted and so they have
strategies
not dissimilar from other rooted
organisms
but then you have mobile
organ linking organisms so it sort of
connects
that all the things that mobile
organisms
like all animals do
and can do with what a factory could do
ecologically
blue
so it just kind of makes me think about
like what kind of biological organisms
take their niche with them
like maybe a snail or a turtle like you
know to take take a large component of
like their habitat essentially
with them wherever they go the mobile
technology made me think of that
yep some people like to travel with uh
light
you know just the keys or something
other people full backpack
every time
stephen

i like this idea of something coming out
of the factory going into the space and
then things being secondhand
a third fourth fifth hand
because there is something there
that is delivered is this
i mean if you're going to use entropy i
suppose you could say
low entropy
kind of nuke new vehicle you know
whatever it is and slowly it gets
incorporated into the ecosystem it
starts to get scratches it starts to get
this and you see in india you know
people start decorating the vehicle you
know the
all this sort of stuff will happen over
time
as
it becomes
somehow incorporated into the niche i
suppose you could argue
so i think it's quite quite interesting
um
how that maybe starts to
change the dynamic i i'm still not sure
whether entropy is so useful
um
term or whether it's
some other way of looking at a metric
maybe something to do with
free energy minimization
as a as a thing but um
better yeah
one other piece of that table that was
interesting was there was multiple goals
like sustainability and fitness
uh

so
it uh is not a
univariate optimization we're not just
doing reward or utility maximization
we can choose multiple goals and we can
have multiple human goals which is
especially important for
things that you know we care about but
then this one of mutualism is
interesting because a lot of times
mutualism could be seen as like an
outcome like the outcome of these two
bacteria's relationship in the niche is
a commensalism or mutualism or symbiosis
whatever it happens to be but actually
using this idea of the
the outputs like the observation states
of a higher level bayesian model
being like the priors on a lower level
nested model
having a goal of mutualism
is implicit in the lower level model
enacting that mutualism
because you're not accidentally going to
metabolically coordinate through
millions of generations
and so this is kind of an interesting
way where
uh like we have to want to
make it work
because we're not going to accidentally
make it work and get along
but that can be a hyper prior
like a belief about how things ought to
be or how we want or prefer things to be
and then we enact policy along those
lines
and by the way

i do um
um
other
uh notes
it's a by the way in um in another paper
i do point out there are limitations to
movable factories so
i'm not uh
saying that this is a panacea
but it's sort of like
like let's just say um cactus versus
like a desert animal
you know the desert animal they have
different ways of getting water and so
but when we can design the system we can
do
trade-offs
like you could have something that's
stationary but then isn't just holding
all of its own money because its money
can be like on a digital system
so it can have some of the advantages of
a stationary organism from an ecological
perspective
like a tree
but then
like the value of the tree
um could be
protected in some other way because we
have some other affordance
and or
that forest could be de-risked at a
higher level because it's like oh each
one of these is we expect to fail this
many percent of the time or we have
these quality control checks on it
yeah i think
a lot of thought is required about about

these things isn't it it's
i don't think there's one big answer but
um
but at least i think one good thing
about
the energy principle active imprints
that it ties
into action and that ties into
energy
so
this is good
yep when you're doing the thought
experiment
pure uh gedenkin then it's like
can be elucidating you know imagine
you're going towards the speed of light
or imagine you have a a
physical mass of this size
but then there's no constraint which is
again double-edged sword on some of
these
attributes that require action
and um
[Music]
so that's that's an interesting idea
that it ties us to
active states which is pragmatic and
useful but also
interesting and then also one thing on
the affective system and then blue you
can ask anything was just
we've talked also about higher order
variables in organisms being kind of
like affective states
whether they were the valence like
positive or negative or the anxiety or
the relative beliefs of an organism um
so just

how do we design for when there's higher
order
states
maybe even ones that could be considered
affective either for
people or for organisms or i mean
organizations that have this kind of
organismal structure but blue
so interestingly like this is going to
touch on what you just said but also
going back to what you said about
mutualism and the goal like has to be
mutualism
so so i don't know um i really have to
uh contact the author of the um
2020 paper uh sims is his name
how to count biological minds symbiosis
the fep and reciprocal multi-scale
integration
which i found um very fascinating but i
would love to invite them on for a
stream someday uh
the the whole point of that is that
there's like nested
cognition happening at the biological
scale and specifically that paper looked
at the squid and the vibrio fisheries
that like light up the squid in the in
the light organ
and um
you know
in that kind of context as in an
industrial organization there's like
this nested cognition happening
um and where
the point of that was the goal wasn't
necessarily
um mutualism or in that case symbiosis

that that's not necessarily the goal but
but
a bigger grander
organism was constructed out of the
union between
these two organisms and so
perhaps in an industrial scale that's
also something that happens right i mean
there's different
parts to the organization that'll keep
it running and you factor in all the
supply chain things and and so there's
this nested hierarchical thing that
happens and something bigger
that has its own kind of cognition
occurs as a result of this like mutual
uh beneficial relationship
we see the paper
the paper we see the paper that stephen
just put up
yeah there's some
these
[Music]
i thought it said in here so i thought
it said in here somewhere about
yeah these different types of
relationships so
mutualism is beneficial beneficial
um
and
parasitism is
beneficial harmful but in between those
there's beneficial and neutral
and
there are several
in between mutual and para
parasitism there are
several

other states and i just remember that
the one of them is beneficial and
neutral
if i could give a kind of random thought
on that and why the informational sphere
might have a different
entropy dynamic than the physical
like
the electromagnetic spectrum
that can have a lot of positive and
neutral relationships like if someone's
operating on two different wavelengths
and one is only on one wavelength then
they can be like of mutual service to
each other or one could just be totally
non-interacting
because they can be overlapping
in space and time
at once and one person might not even
have to be aware of it
whereas in the tangible
um
i guess most embodied one can be
you know only one thing per spot
that's zero sum there just can't be two
two atoms in the same spot
or
maybe you know two
laws
playing out
in one territory
but informationally there's probably a
lot more multi-channeling
and that might give enough room
for practical success if not ultimate
theoretical answerings
stephen slit
and then blue

this brings up a lot of issues it's very
fascinating um
i'm i'm curious um
so coming back to this idea and i i mean
daniel mentioned hyper priors i things
that we kind of have fixed um
i actually looked that up actually to
make sure i understood it but
things that we have is very distinctive
and i'm again trying to compare these
um
these types of um
environments when one where you've got
an agent wanting
or being able to just
calibrate in amongst the noise and ones
where you want to just keep it as clean
as possible
you know so
and and the challenge is sometimes when
you have a
quality management system in those
places so for instance i suppose one
thing that comes to mind is the
experiment with a rat that they used to
have and they've had for years about
having cocaine or addictive substances
and how
they would they only needed to have a
couple of times and they would be
addicted and it was seen as this kind of
proof of principle about addiction and
how it works and then the more recent
work where they instead of putting them
in a bare box
and just having the cocaine or whatever
they were being
um sort of

exposed to they were in a more
ecological
um
boxed you know where there was things to
do and they suddenly found that the
behavior of the animal was very
different they wouldn't necessarily
become addicted in the same way
because they would actually because they
were in a better ecology
now in theory the entropy's gone up
there if you're looking entropically
but for the animal they've been able to
engage and and dynamically process that
that's very different to
how you would try and
create you know a medical
production line right you you want to
keep it clean you want to keep it
sterile
um you want to have those sorts of
things
so i'm just curious about that that and
then the danger maybe in the biological
system of this hyper prior say the
some sort of kick or something that
comes from food
not having any other compliments around
like like as daniel you know so there
isn't other things so that just
naturally kicks in like the reward goal
which may be always there as a default
if there's not other things to
complement it it just it starts running
the show and that could be part of
how we get into these maladjusted
behaviors that you were talking about
i wonder what your thoughts are on that

just quick point of clarification that's
not what a hyper priority is that's not
a hyper-stimulating stimuli a hyperprior
is a is a prior over another prior in a
bayesian
multi-level framework so it doesn't have
to be accentuating or differentiating or
extreme or maladaptive it's a general
framework for the relationship between
variables
yeah well i was thinking it was more the
hyper prior could give a it's a it's a
fairly fixed priority and it's one which
is a go-to solid prior and it's useful
for being that does just saying having
that present hyper fire could be fixed
or learned
or learned
okay okay
but it tends to be hyper prior means
it's quite accurate isn't it it's quite
it's not fuzzy hyper is i think a
spatial designation like supra
not
not in the extreme case hyper is only
referring to the fact that it's a a
parameter about
another parameter
so it doesn't tend to be hot or cold or
anything it's just the general phrasing
okay got you yeah
so but stephen fox if you want to add
anything
otherwise
blue
okay we see the head
yeah so
in

now the effective system
so
here
here's my diagram
so in
in this um
saying that uh
it's a little bit so small i can't see
it myself
um
but if there's the same
starting point
but um
of a generative model so if the human
being
in human robot
interaction knows that um
okay so i i'm i'm in the predictor i'm
human i'm in the predictive processing
business that's what i'm doing and um
and
um so i've got prior beliefs
and i have got
generative models
then
the
[Music]
robot
can be
um what does that say
personas and scenarios for robot based
on the humans uh
well that's world view but generative
model
and human readable model of the robot
mind that corresponds to that
so
that's um

and this is in
aligning human psychomotor
characteristics with robot an augmented
reality
in this
so
there's a bit of a model there
that's also very related to the felix
shoulders paper
trust us extended control active
inference and user feedback during human
robot collaboration
ooh
i'll put it in the youtube channel
i'll put in youtube chat
but brilliant go for this one
so that's um
it's all in the past experiences
physiology so this is body memory in
there personality gender culture emotion
reasoning in biocybernetics
so this paper i'm relating that to what
how this can affect interact with robots
exoskeletons and augmented reality
there it is
we got here a few equations
there it is but
here the two
i point out in this that um
you find
yeah so
in previous production literature the
uncertainty of human behavior
this is this is true
so in previous production literature the
uncertainty of human behavior has been
recognized as a source of productivity
quality and safety problems

however fundamental reasons for the
uncertainty of human
behavior have been framed
previously
as a black box
so what i was trying to do there was
relate it to
to open up the black box
and
[Music]
frame it as an open box
and then
fundamental reasons can be aligned with
production technology to facilitate
improved production
performance
so that's as far as i've got with that
it's almost like for the
reward utility driven person you can say
there's going to be an increased
expectation of performance
for the uncertainty minded you can say
there'll be a reduced variance
for the resilience minded
or for the reconfiguration minded that
there'd be an increased capacity
to rapidly reorganize
so i think that
we'll have all or none
blue
so my thoughts kind of dated at this
point but just going back to um
touching back on to mutualism and
um you know these these the
electromagnetic spectrum when you're
talking about how like you know the
signal can amplify
um

there's also like the possibility for
the signal to cancel right if if the you
know wavelengths are are the same and
they're heading in opposite directions
it they just directly cancel each other
out and it kind of made me think of like
cancel culture and also like the
relationship between um
you know the parasite is is also like a
nested biological relationship um even
though it's not necessarily beneficial
but there's still this nested layer like
the parasite is unable to function
without the host of course
but um
the host embodying the parasite like is
a completely different
cognitive cognating creature and i think
about like the zombie ants right that um
i don't know if you guys have heard
daniel can probably unpack the zombie a
lot a lot better than i can but it's
like a parasite that then drives the
behavior of of the ant due to some like
rewiring of the cognitive process
so even in in the parasite instance
there's still a greater cognizing
structure
and
going back to neutralism again the
the other terms are so commensalism
is where
one gains benefit
while the others
it hasn't there's a new neutral no
benefit no harm
and then there's a mentalism where one
is harmed or the other is unaffected

so they're in between so there's first
mutualism
benefit benefit then commensalism
where there is uh benefit
neutral
and then there's a mentalism where
there's um
harm
and neutral
and then parasitism where there's one is
harmed and the others benefits
i i wonder if this has to do with the
movement from maybe just its semantic
but from sustainability to regenerative
because sustainability it's like the
sustain
on the guitar sound of the bell like
let's sustain
let's have this car and sustain it or
make it it
can be meant to mean more than that but
it's about
sustaining something that exists
versus regenerating
uh
means like we're gonna have to be doing
more than just
persisting or stabilizing it
that it has to be also actively having
this process
of uh
of regeneration which always involves
like multiple parts like bucky fuller
saying that unity is plural and at
minimum two
so then now thinking about how
from polyculture of growing crops
to monoculture

the green revolution and everything
and then now oh maybe there should be
the fish in between the roots
of the plants and then the birds and the
mushrooms like this
are these stable ecosystems that is what
remains to be seen but some human
modified
niches like sourdough bread and things
and beer have been really successful so
maybe there's higher order combinations
too that could be really successful and
be
very productive but
um
this is kind of a framework that helps
us at least explore that so that's quite
cool
i was at um
remotely of course it was there was a
launch the south african um
grassroots innovation programme they
did um
a launch
yesterday with five
uh
lady innovators and one of them was
doing um
movable aquaponics
[Music]
she seemed to have addressed
it's quite a sophisticated system
but it's all uh
it's all set up in buckets that can be
carried
one person can carry the buckets that's
the size they are
and then you build up

the
the farm by different
just having the
number of the different the more buckets
you get the bigger the aquaponic farm
so that was interesting
and then again with the physical zero
sum and then the
electromagnetic or informational or
financial non-zero sum
um
if the overlapping is just
given the tools that already exist
in the ecosystem that you find yourself
in like your dropped on survivor island
you still may be able to find some
synergies most of them you won't know
but they're over long time traditional
ecological knowledge can learn about
some medicinal combinations and
oh these two should be combined for the
boat
and these kinds of things but
design
and this is related to what steven said
about the
product being plugged into
another external niche and what you
wrote about the cupboard being attached
to the wall like we could design some
pretty minimal pieces
that might really scaffold
you build a trellis and then something
grows within it
so the engineering might not be
huge
to connect a few pieces
that have a

latent capacity to work together
they the word you use their scaffolding
has that got any
um
of course i've heard it before but
does that have any
[Music]
basis in nature to your knowledge
i would certainly say so um from the
spicule structures like in bones
and the way that um physical structures
are scaffolded to um
embryogenesis and the developmental
scaffold scaffolding of
the uterus and
the
core caretakers and
different insects and humans
and the family and so i think there's
the cultural scaffolding
and then all the way down to the
microstructural
cytoskeletal scaffold
for scaffolding enzyme complexes that
then can do very efficient things too
yeah it's interesting because the
problem with
the the challenge
with um so-called grassroots
entrepreneurship and innovation
is scaling it up
that's the the challenge that's
that's what was said yesterday
but that's what said in india
there's this problem that
solutions are developed
and they they work
but they at best stay where they've been

in the village where it's
been developed it sort of just stays
there
and so
scaling up and it's a um
open source um
would it's
well there it is the technology is there
isn't an open source ecology it's uh
it's possible to put all engineering
designs and builds and materials
and
um
[Music]
methods of work and
all on the internet where anybody can
access it but still
things are not scaling up
thanks blue
so i think about scaffolding um in terms
of
like neural systems but also in societal
systems
in the brain like my phd work was
focused on
finding a scaffold that will
promote the growth of neurons
across um
across the lesion
and so when you have a traumatic brain
injury the glia which is like the
natural scaffold like freak out and form
a scar to prevent like the whole rest of
the brain from dying because cell death
can be like this kind of cascading
process and so so the glia and and it
literally results in like a hole in the
brain where the injury was

so i was looking at these bioengineered
scaffolds that promote the growth of
neurons across that hole so so like
it'll it'll form a bridge or a structure
scaffold for the neurons to grow
um and then you know in terms of the the
societal um setting
i think about um scaffolding as
infrastructure
and and i mean even in an organization
like you have to have
bridges and roads and internet
connections and utilities and you know
these things are it's the the
infrastructure that's necessary for
kind of new
um
branches or systems start up and even in
like the farming
terms it's like you know the the um like
i don't know here we just call them the
ditches but like the irrigation canals i
guess is probably the proper term like
that you know the um where you can
get a river to flow off into different
subsections and then irrigate the
farmland with that water so so i think
about that as as the scaffold
awesome blue nice connection and sounds
like cool phd research we see your
screen stephen
so you see grassroots innovation program
yes
okay yeah so that was that was yesterday
um although the press release was
in july but it's not it's not detailing
the book but there we are this is the um
so in south africa there were

the rate of unemployment is
i think officially it's
more than 30 percent
but um unofficially it could be as high
as 50
and uh
this is
so they're now trying to address
at least partly through this grassroots
innovation problem
program
so one and there are there are similar
and we know that
um
the book decades ago small is beautiful
and
gandhi was
before that was promoting this but
it does it's not scaling
and so there are these
in between industrial production there's
just this ever-increasing wasteland
but efforts are going on around the
world
and grassroots is a good term because
it's just growing from
evolving naturally doesn't it
sounds like really
interesting work thanks for sharing it
stephen celet
yeah actually that ties in i actually
was running a project in south africa
funny enough
um and working on
looking at how to create community
driven innovation through strategic
strategic conversations at the community
level and there is there is an

interesting difference between
grassroots i think and community driven
so i think that grassroots is often kind
of um
tied to
it's like the base level
of the organizational structures you
often hear it with political campaigns
and stuff is the grassroots but those
grassroots are the grassroots of a
campaign where people sort of get into
there which is a little bit different to
the kind of a community embedded
community driven
scale and that comes back to the scaling
question which is the same thing that
had the problems that so they're scaling
there's now when you talk about scaling
up and that's a big
challenge is how does stuff get scaled
up and they're often now talking as well
about
scaling out
if you have a program or scaling deep
so
you know some things might scale up
because they go up to a higher level and
then they basically get dropped down
again
but how do these things get scaled out
so
there may be some i think you're you're
you're tapping into a few
multi-level issues which i think are
quite quite important so i wonder what
your thoughts are about
going at uh
going into these kind of

embedded local
context to create
new seeds so to speak for like
adaptability
and innovation
it's like a seed is a biological
planting metaphor the scaffold is a um
highlighting the industrial component of
a biological process
like you still need the tomato seed so
it's not saying it's not community
driven it's just like a soil metaphor
like how these
communicate what our vision is and they
really do matter
and so they're really um
important points
actually here's a comment from kevin in
the chat who wrote
is the scaffolding a schema or is this
supposed to be anatomical
that is is there some physiological
analog in the human nervous systems or
genome that corresponds to the
scaffolding i think blue made an awesome
point about the glia and about
connective tissue being
an immune tissue being scaffolding
physically for like the synapse and for
cells and then the genome oh yes
the chromatin scaffolding goes deep in
the nuclear pore structure
and that's a whole cool area and
maybe there'll be a day for
transcription factor active inference
soon enough
so um what you're saying these hands
this is a an organization um

active in botswana
and everything they're doing there
but
there we are you can see
yep
[Music]
yeah
so this is led by uh
thurbison
and
he's um applying design methods in um
uh communities in southern africa with
great effect
and this is all uh
and
so the idea is that people are
with their own hands are
developing
innovations and
as you say you can be scaled out
[Music]
for example in
in that part of africa that
there are many
movements like uh
many initiatives going on
but it would be very beneficial if there
was um
a
framing
a kind of unifying framing of what
they're all doing
which can be seen
as an alternative an alternative
form of
socio-economic development
rather than just as something that's uh
well fundamentally

inferior
to industrialized to
to the conventional model of
industrialization
so all suggestions are welcome
i just put in the youtube chat a volume
uh
by capparel grissomer and wimsat
developing scaffolds in evolution
culture and cognition
and so the paradigm change as steven
kind of hinted at very early is like
from systems where we eradicate the life
to where we
welcome the life and all that it brings
and we don't need to build that and we
can't build it we're not going to be
able to compose the lipids
and the millions of years of fault
tolerances into some sort of uh
you know
monster that
helps our living systems
we don't even have to though we just
have to build an engineered scaffold and
the scaffold can be
radically different than what
the final product looks like the
scaffold can even disintegrate if it's
biodegradable
so
it um it's a really interesting area
that this discussion went which was like
towards
the engineering that
makes us think of like foundries
and some of the technical and precision
processes

tying them to some natural processes
like this evolution culture cognition
volume does
and unifying frameworks
those uh
together almost point a way towards
engineering a scaffold not a booster
rocket
or just a bunker
but bunkers might come into play of
doing some
scaffold design
so that there could be
reduction of expected free energy
through deep time
so pretty interesting um point stephen
thanks for sharing all these resources
too blue
just a quick comment um you were talking
about the scaffolding disintegrating
sometimes the scaffolding is designed to
disintegrate like i think about in in
biological you know terms we have like
webbing between our fingers that enable
like the growth of the digits and then
you know it's designed to be eroded uh
in in the developmental process um so
sometimes that's that's part of it or
even like when you get a wound sealed
you know you have like the the stitches
that scaffold that form the scaffolding
so the healing process can occur but
they dissolve inside the wound
um
actually that's a good that's quite cool
to think about that as well in the
processes
and uh when when we

whether in areas that we live in or
areas that we don't live in when we plan
to intervene are we planning to make
something that sustains itself
or are we planning to
deliver or co-create a scaffold that
is planned to disappear
those are really different views on
local and distal intervention
steven with a screecher
what's on the screen stephen fox yes
so now this is um
this is an initiative called connected
hubs
so this this is coming from the southern
african uh innovation support program
and
uh the idea is to
um
that there are
hubs
which support
[Music]
innovation and entrepreneurship
and these are
this particular
one is um
so as you can see the connectivity is
here
in this part of africa
and what we do is build capacities and
support
and uh
so it's connecting uh different
hubs together
and so a hub it's just
what that means is it's just
that usually they've got a physical

location
and they
uh
there are people you can contact there
remotely and physically
you've got
can help with developing um
ideas for entrepreneurship and social
entrepreneurship so this is something
that's going on
it's connecting hub
so maybe there's
well
this is
hopefully this is the intention is this
persists after the southern african
innovation support program is on the
second phase now after it finishes
so this is one
i suppose it's summed up in this uh
diagram here this idea that there are
more and more
hubs herbsing
and the european commission is um
trying this uh
africa
eu digital
innovation
hubs oh we'll see this and then
there we are so this is kind of things
that are going on so i'm involved in
this project sustainable
um network of digital innovation hubs
so that's something that's uh
coming down from the african union and
the european union
and the hope the aim is that
this is an uh a somewhat new

way of
facilitating socioeconomic development
and the idea is to try to get auto
catalytic effects by you know more
connections
more um
more dynamism
well they're always welcome to
come talk about active inference at
whatever stage
in the last few minutes we'll just have
steven and then blue
and
yeah just give kind of last
thought or question here as we close out
okay yeah no thanks very interesting and
um
i suppose with active inference i think
very interested in these these
developments it's actually how i ended
up getting into this because i was
trying to develop community-based hubs
in south africa funny enough in rural
rural maputo land
um and i think what's quite interesting
is daniel's this this idea of a unifying
framework rather than trying to reduce
down all the complexity which is what i
was doing for far too many years
is really amazing and the other thing
that all of these
aspects that you're taught you've just
shared and that ties in with this lab is
it's it's about participation as soon as
you bring in participatory processes
you
you bring in these active inference
dynamics you bring in this complexity

you bring in this need
but also the ability
to do what
mass
manufacturing can't do
but like i thought you said it's it's
seen as being
inferior or in our culture and i think
seeing it as being
vital but also challenging is really
cool so i think that's something i'm
going to take away because i can't i
won't actually get to stay too much
longer i've got to jump on another call
but that's something i'd like to take
away so thanks very much brilliant
really really enjoyed today thank you so
much
thank you steven
see you later
blue
um so just a couple final thoughts uh
john boyk and his idea of connected hubs
i mean he was on like probably six
different um live streams
and uh he gave a very good overview of
sustainability um and the you know
how to build an architecture toward a
sustainable future
um so the connected hubs kind of
reminded me of that and then
um the one other thing was um just about
scaffolding and
you know how do we build kind of a
scaffolding for our cognitive
architecture and is this a scaffolding
that's maybe designed to dissolve and so
i think about you know scaffolding for

our cognitive architecture as as
education right like we send the kids to
k-12 and they you know form this like
shared understanding of reality and and
two plus two is for and how to read
books and so forth and
this kind of um education maybe daniel
will remember at least in college when i
started college um in biology many
things are so complicated that they just
give you like a very broad overview that
doesn't make sense or i was always
searching for like but that doesn't make
sense but that doesn't make sense like
in your entry-level classes and and then
as you get like deeper and deeper into
studying biology or biochemistry you
start to kind of understand electron
transport and these different like
things that are are very like micro but
the macro level doesn't make sense
without understanding that that micro
and in my mind this this kind of
educational or that way that i was
taught in college was kind of like
building a scaffold that was designed to
disappear
anyway
um but anyway yeah thanks for the for
the talks today and for coming on it was
good
if i can get my last comment and then to
the author for the final comment
i just love this idea of the
biochemistry education as the removable
scaffold like you learn it once that
there's the essential amino acids and
then you never wonder like why on the

nutrition label is there essential and
non-essential you forget the details the
side chains you know the c minus on the
test that all disappears
with something that's generalized but
otherwise we would look at macroscopic
phenomena like why can't i swim when
there's this algae in the lake
well there's some molecular answer we
could know at least how to sense make in
that area might even know the specific
details but also it can be removed in a
way
but can it be removed
intergenerationally
a chemistry to alchemy reversion
with similar or different tools
that would be quite something
so stephen
with the um final notes but just wanted
to say thanks for coming on these
streams and it was just like super
interesting discussions
well thank you very much it was
my pleasure and
i look forward to
further
cooperation with you all
great if you or any of your
collaborators on these
projects or your contacts these hubs
wanted to
present on this or
facilitate in any way then we can talk
um another time
very good thank you very much
thanks again steven thank you blue bye
you